Ahmedabad University

Weekly Report 4

Disease Prediction System using Machine Learning

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Course: CSE 620 Machine Learning

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Week 4 Feature Preparation and Dataset Splitting

Work Completed

During this phase, the dataset was prepared for machine learning model development by organising the features and target variables required for classification.

The symptom attributes in the dataset served as the input feature set (X), while the disease column was designated as the target variable (y) that the model would learn to predict. This separation ensures that the machine learning algorithms clearly distinguish between input variables and prediction outcomes.

After this step, the dataset was split into training and test sets using the train_test_split function from the Scikit-learn library. This separation ensures the model can later be evaluated on unseen data, helping assess its ability to generalise beyond the training dataset.

Tools / Technologies Used

- Python
- Pandas
- Scikit-learn

Challenges Faced

Ensuring the dataset is properly structured and correctly split into training and test sets, so evaluation results accurately reflect the model's predictive performance.

Plan for Next Week

- Implement baseline machine learning models.
- Train initial algorithms such as Naive Bayes and Decision Tree classifiers.